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General CNN model for biomedical image classification via genetic algorithm-based hyperparameter optimization

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ABSTRACT

This study addresses the challenge of hyperparameter selection, a key factor affecting convolutional neural networks (CNNs) performance in biomedical image classification. A genetic algorithm (GA) is employed to optimize activation function, padding, number of filters, kernel size, dropout rate, pooling size, and batch size. The optimized CNN is trained on brain Magnetic Resonance Imaging (MRI) images of Multiple Sclerosis (MS) and validated on Alzheimer's MRI and COVID-19 chest X-ray datasets. Results show substantial improvements across all datasets. On MS, the proposed model achieves up to 37.6 % F1-score and 33.7 % accuracy gains compared to other models. On Alzheimer's, improvements reach 32.8 % in F1-score and 32.5 % in accuracy. For COVID-19, gains are smaller but consistent, ranging from 0.8 % to 12.1 %. Overall, the GA-optimized CNN consistently outperforms widely used architectures such as Xception, InceptionV3, VGG16, VGG19, AlexNet, ResNet50, and GoogleNet, demonstrating both enhanced classification performance and strong generalizability across biomedical imaging tasks.

1. Introduction

Biomedical image classification has been studied for many years as a field that plays a critical role in medical diagnosis, disease detection, and biological research. Although traditional image processing and machine learning methods that manually extract specific features perform well on some data sets, their limitations in handling complex biomedical data have become apparent over time. Especially in tissue, cell or organ images, capturing fine details and making accurate classifications have not always been possible with classical techniques. However, in recent years, it has been observed that deep learning techniques, especially Convolutional Neural Networks (CNNs) [1], have made revolutionary progress in this field. CNN can achieve high accuracy rates without human intervention by autonomously learning meaningful and distinctive features from raw images [2]. This development has improved the effectiveness of numerous biomedical applications, including cancer detection, radiographic and magnetic resonance imaging, and cellular imaging [3,4].

The model's performance mostly depends on the network hyperparameters, such as the architecture, optimization techniques, the number of layers, filter sizes, pooling methods, the optimization algorithm used, initial weights, activation functions, and learning rate. Nevertheless, no

mathematical formula or rule can be used to ascertain the hyperparameters of the models. Therefore, the model's design and settings should be carefully selected, and application-specific optimization should be performed to achieve the best results in CNN-based applications.

Pre-trained and proven successful architectures such as AlexNet [5], VGG [6], and ResNet [7] are frequently preferred in image classification and object detection tasks. AlexNet pioneered the popularity of deep learning and achieved high accuracy on large datasets. VGG provides detailed feature extraction using small filters and deeper structures. ResNet allowed the training of much deeper networks by using residual connections to solve the vanishing gradient problem encountered in deep networks and has gained an important place in image analysis. In addition, newer, more advanced CNN-based architectures are widely used in the biomedical field. While DenseNet provides a more efficient feature learning process by increasing information flow through dense connections between layers [8], InceptionNet offers a flexible structure by simultaneously extracting multi-scale features using filters of different sizes [9]. EfficientNet offers an efficient approach, achieving high accuracy with a structure that requires less computational cost. Smaller architectures such as MobileNet and ShuffleNet are preferred for image analysis on mobile devices or embedded systems [10]. These architectures offer advantages for specific data types and application scenarios;

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Abbreviations

ANN	Artificial Neural Network
BBO	Biogeography-Based Optimization
CNN	Convolutional Neural Network
COVID-19	Coronavirus Disease 2019
F1-score	Harmonic mean of Precision and Recall
FN	False Negative
FP	False Positive

GA	Genetic Algorithms
GWO	Gray Wolf Optimization
KNN	K-Nearest Neighbor
MRI	Magnetic Resonance Imaging
MS	Multiple Sclerosis
PSO	Particle Swarm Optimization
TN	True Negative
TP	True Positive
WOA	Whale Optimization Algorithm

therefore, proper architecture selection directly affects classification performance.

Although these architectures are frequently applied to achieve successful results on image data, they have some important limitations. First of all, they are usually designed manually for image datasets, which limits their direct generalizability to different types, such as biomedical images. Additionally, as the complexity of deep architectures increases, the parameter space expands, making it more difficult to train the model.

Hyperparameters, such as the number of layers, learning rate, activation functions, filter sizes, and optimization algorithms, directly impact the performance of CNN models [11]. There are many potential values or options for each hyperparameter. Therefore, finding the optimal combination using traditional trial-and-error methods is computationally expensive. The use of metaheuristic optimization techniques such as genetic algorithms (GA), particle swarm optimization (PSO), and differential evolution might be a good option to determine the optimal hyperparameters. These techniques are particularly effective at improving model performance and reducing computational costs of hyperparameter optimization due to their outstanding ability to search within the search space without trying all possible options.

GA is one of the most efficient and frequently used search techniques. Since GA can traverse the search space faster than trial-and-error methods and optimize the combination of parameters, using GA may provide an advantage in determining CNN parameters, thus improving CNN performance by selecting optimal or near-optimal parameters [12]. This study employs GA to optimize CNN hyperparameters and aims to find the answer to the following research questions: (1) Can GA improve CNN performance in biomedical image classification by optimizing network hyperparameters, and (2) Does the GA-optimized CNN model demonstrate generalization capability across unseen biomedical and non-biomedical datasets?

More specifically, first GA optimizes CNN parameters, including padding, number of filters, kernel size, pooling size, activation function, dropout rate, and batch size, considering the model's classification performance on the Multiple Sclerosis (MS) dataset consisting of Magnetic Resonance Imaging (MRI) images [13]. Subsequently, the generalization ability of the CNN with the optimized hyperparameters is evaluated not only on two additional biomedical datasets, the Alzheimer MRI dataset [14] and the COVID-19 chest X-ray dataset [15], but also on a non-biomedical benchmark, the CIFAR-10 natural image dataset. The performance of the optimized CNN on these datasets is compared with the performance obtained with other state-of-the-art CNN architectures, including Xception [16], VGG16, AlexNet, ResNet50, InceptionV3 [17], VGG19 and GoogleNet [9]. Experimental results demonstrate that the GA-optimized CNN consistently outperforms these well-known models across all biomedical datasets and achieves competitive performance comparable to ResNet50 on the CIFAR-10 dataset, despite being optimized for biomedical images.

This study introduces a systematic and adaptive framework that employs GAs to optimize CNN hyperparameters for biomedical image classification tasks. In contrast to prior works that primarily apply predefined CNN architectures to specific datasets, the proposed method

aims to automatically configure critical network components through an evolutionary search process guided by classification performance. By conducting comprehensive evaluations on multiple biomedical imaging datasets with differing modalities and diagnostic targets, the study demonstrates that the GA-optimized CNN achieves superior classification accuracy and exhibits strong generalization capabilities. In addition, an experiment on the CIFAR-10 dataset confirms that the proposed model maintains competitive performance even beyond the biomedical domain. These findings indicate that GAs can effectively optimize CNN hyperparameters, resulting in configurations that deliver both high accuracy and robust generalization across diverse image classification tasks. Consequently, this work contributes to the development of more adaptable and efficient deep learning pipelines, addressing key limitations of manual design and dataset-specific tuning.

The key contributions of this study can be summarized as follows:

- GA-based CNN hyperparameter optimization: We propose a systematic framework that applies a GA to optimize critical CNN hyperparameters, including activation function, padding, number of filters, kernel size, dropout rate, pooling size, and batch size.
- Cross-dataset generalization: Unlike many prior works that focus on a single dataset, the proposed GA-optimized CNN is trained on the MS MRI dataset and tested for generalizability across multiple datasets and domains, including two biomedical datasets (Alzheimer's MRI and COVID-19 chest X-ray images) and one natural image dataset (CIFAR-10). This evaluation demonstrates that the optimized architecture not only generalizes well across different biomedical datasets but also achieves strong performance on a natural image dataset (CIFAR-10), performing comparably to deep architectures such as ResNet50.
- Performance improvements over state-of-the-art models: The GA-optimized CNN consistently outperforms widely used deep learning architectures, including Xception, VGG16, AlexNet, ResNet50, InceptionV3, VGG19, and GoogleNet. Experimental results show up to 37.6% F1-score gain and 33.7% accuracy gain on MS, 32.8% F1-score gain and 32.5% accuracy gain on Alzheimer's, and consistent improvements of 0.8%–12.1% on COVID-19. Furthermore, cross-domain evaluation on CIFAR-10 demonstrates that the optimized architecture maintains competitive performance compared to ResNet50, indicating its generalization capability beyond the biomedical domain.

The remainder of this paper is organized as follows: [Section 2](#) presents a review of the existing literature. [Section 3](#) provides the preliminary background, including CNN components, the GA system, and the datasets used in the study. In [Section 4](#), the optimization of CNN hyperparameters using GA is described, followed by details of the GA system in [Section 5](#). [Section 6](#) reports the experimental results, including the performance evaluation, sensitivity analysis, and cross-domain evaluation on the CIFAR-10 dataset, along with a discussion of the findings. Finally, [Section 7](#) offers a general assessment of the findings and discusses potential directions for future research.

2. Related works

In the literature, deep learning architectures such as AlexNet, VGG16, VGG19, and GoogleNet have been extensively utilized for classification, segmentation, feature extraction, feature selection, and related tasks [18,19]. In addition to deep learning algorithms, various studies have incorporated optimization techniques [20,21]. This section examines applications incorporating neural networks, including deep learning and optimization algorithms, for biomedical image analysis.

Zhou and Shen [22] optimized the weights and biases of a feedforward neural network to identify multiple sclerosis lesions in MRI images using the Biogeography-Based Optimization (BBO) algorithm. Similarly, Albadr et al. [23] proposed using the PSO algorithm to optimize the weights and biases of an extreme learning mechanism, which is a form of feedforward neural network for COVID-19 classification. Another study, Salem et al. [24], also applied PSO for hyperparameter tuning in various machine learning models, including Boosted Random Forest, Decision Tree, Support Vector Machine, K-Nearest Neighbor (KNN), Multilayer Perceptron, and Artificial Neural Networks (ANN) for COVID-19 classification.

Lima [25] proposes a method based on wavelet entropy and PSO for MS disease detection. Wavelet entropy is used to extract image features. The PSO algorithm is used to optimize the weights and bias values of the neural network model. The method is tested on a small dataset comprising MRI images from 38 MS patients and 26 healthy individuals.

Dhakshinamoorthy et al. [20] propose a hybrid approach combining the Whale Optimization Algorithm (WOA) and the Gray Wolf Optimization (GWO) method to segment brain subregions in Alzheimer's disease. Then, the AlexNet fed by the segmented images classifies Alzheimer's disease. Lu et al. [26] proposed the KNN model based on the sparrow swarm algorithm for early diagnosis of Alzheimer's disease. In this study, the K value of the KNN algorithm was optimized, and the test was conducted across five datasets.

Other researchers [21] propose a CNN model to extract deep features from both computed tomography scans and X-ray images for COVID-19 detection. The proposed model applies sparse filtering, feature transformation, particle swarm optimization, GWO, Hybrid WOA and entropy mutual information techniques to the extracted features by a CNN network to select essential features, then applies several feature fusion techniques to fuse the selected features and finally applies fully connected layers to classify the input images.

Although these studies [20–24,26] achieve good performance, they are data-specific studies; therefore, when the dataset changes, they need to be re-run, and the corresponding parameters need to be optimized again.

Das and Saha [27] optimize the CNN architecture with GA for diabetic retinopathy. This study optimizes the number of convolution layers, the number of kernels, and the kernel size of convolution layers. Similarly, Salih and Duffy [28] used a GA to optimize some of the hyperparameters of a CNN model applied to skin lesion classification data. In the study, GA aims to optimize the momentum, epochs, activation functions, dropout, network depth, batch size, learning rate, number of filters, number of nodes, and dense layers. The optimized

CNN is tested on four other datasets of skin lesion images. The proposed algorithms [27,28] succeed with high accuracy thanks to GA optimizing the CNN hyperparameters. Aguerchi et al. [29] optimized CNN's kernel size, stride, and filter number parameters for breast cancer classification with PSO. The authors report that the model is not generalizable for different datasets, even if two datasets are used for evaluation. With the same purpose, Ait Amou et al. [30] and Atteia et al. [31] employ Bayesian optimization for different data, brain tumor classification and diabetic maculopathy diagnosis, respectively.

Asiri et al. [32] optimize various parameters of a CNN for brain tumor diagnosis, including the number and size of filters, stride, padding, pooling techniques, activation functions, learning rate, batch size, and the number of layers. Similarly, Aamir et al. [33] and Shivwanshi and Nirala [34] optimize some of the CNN's hyperparameters. In these studies, the authors determine the variable range for each parameter and manually select the values that yield the highest performance.

In addition, Armoogum et al. [35] investigated breast cancer prediction using transfer learning-based models, Kuklin et al. [36] proposed a technique for segmentation and selection of biomedical images, and Nobel et al. [37] introduced an advanced segmentation model (ResECA-U-Net) for gastrointestinal cancer radiation therapy planning. These works further highlight the growing diversity of deep learning and optimization strategies in biomedical imaging.

Although existing studies demonstrate promising results, most optimize CNN parameters specifically for the given dataset, and their generalization capability has rarely been tested on other datasets. Consequently, these approaches overlook the potential influence of dataset characteristics or modality variations on performance. This study addresses this important limitation by introducing a generalized CNN framework optimized using a GA and evaluated across multiple medical imaging modalities.

Table 1 provides a comparative summary of the reviewed studies. As shown, the majority of existing works employ optimization techniques such as GA, PSO, WOA, or GWO to fine-tune neural networks or machine learning models for dataset-specific tasks. While these approaches achieve high accuracy, their applicability remains constrained to the datasets used, and their generalization across different imaging modalities has not been systematically validated. In this study, we address this limitation by presenting a GA-optimized CNN framework evaluated on diverse biomedical datasets (MS, Alzheimer's, and COVID-19), thereby demonstrating its robustness and broader applicability beyond single-dataset scenarios.

3. Preliminary

3.1. Convolutional neural networks

CNNs are widely used artificial neural networks, especially in image processing and deep learning [38]. A CNN basically consists of three blocks: convolution, pooling, and fully connected layers. While convolution layers use trainable filters to detect local features in the input data, pooling layers reduce the computational costs by reducing the dimensionality of the input matrix, and fully connected layers perform

Table 1
Comparative summary of related works.

Study	Optimization method	Application / Dataset	Limitation
Zhou and Shen [22]	BBO	MS lesion detection (MRI)	Dataset-specific
Albadr et al. [23]	PSO	COVID-19 classification	Dataset-specific
Salem et al. [24]	PSO	COVID-19 classification (various machine learning models)	Dataset-specific
Lima [25]	PSO + Wavelet entropy	MS detection (MRI, 38 patients, 26 controls)	Small dataset
Dhakshinamoorthy et al. [20]	WOA + GWO	Alzheimer's segmentation + AlexNet classification	Dataset-specific
Lu et al. [26]	Sarp Swarm	Alzheimer's diagnosis (KNN)	Dataset-specific
Das and Saha [27]	GA	Diabetic retinopathy (CNN optimization)	Dataset-specific
Salih and Duffy [28]	GA	Skin lesion classification (CNN optimization)	Dataset-specific
Aguerchi et al. [29]	PSO	Breast cancer classification	Limited generalization

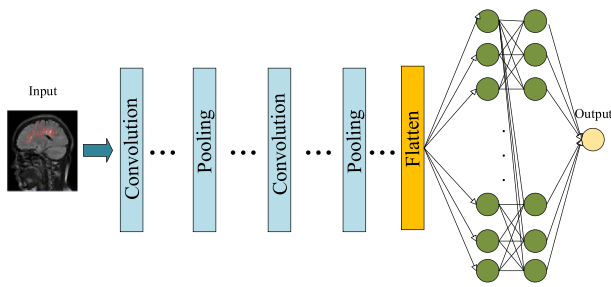


Fig. 1. An example CNN architecture.

the final classification using the extracted features through the convolution and pooling layers [39]. A CNN typically starts with convolutional layers and ends with fully connected layers. However, the number of convolution and pooling layers, as well as the size of the fully connected layers, may vary depending on the design preferences and requirements of the user. Besides these components, CNN also has some additional hyperparameters, such as padding, activation function, dropout rate, and batch size, which need to be determined. An example CNN architecture is shown in Fig. 1.

3.1.1. Convolution layer

The convolution layer uses filters (which are usually trainable) to detect and represent local features in the input data. This layer performs convolution operations on the input image using a sliding window method with small-sized kernels [40]. Each convolution operation generates feature maps by calculating the weighted sum of specific regions in the input data. The number of filters and the kernel size in each convolution layer depend on the user experience [41]. The convolution layer increases the learning capacity of deep networks by making high-dimensional visual data more meaningful and easier to process.

3.1.2. Pooling layer

The pooling layer reduces the spatial dimensionality while preserving important features [42]. This layer summarizes the input feature maps in a certain window size by sliding the window. One of the most common methods is max pooling, which highlights important features by selecting the largest value within each window. Average pooling, on the other hand, performs a smoother summarization by averaging the values within the window.

3.1.3. Fully connected layer

The fully connected layers are at the end of the CNN architecture and consist of layers where each neuron is connected to pre- and post-synaptic neurons [41]. They are fed by the feature maps and produce the final outputs for tasks such as classification or regression.

3.2. Genetic algorithm

GA is a population-based evolutionary computation technique inspired by biological evolution processes. It searches for solutions in the search space using genetic mechanisms and the natural selection principle [43]. GA initially generates a population randomly, and each candidate solution, called a chromosome, is evaluated by a fitness function. The next generation is then created by applying the genetic operators: selection, crossover, and mutation. In the selection phase, the higher the fitness value, the greater the chance of participating in the reproduction process for chromosomes. The crossover process increases diversity by providing gene exchange between the selected individuals. Mutation prevents the population from being trapped in local minima by introducing small random changes to the genes of the chromosomes. This process continues until a stopping criterion is reached. A flowchart for a basic GA is shown in Fig. 2.

3.3. Datasets

In this study, we use three biomedical datasets: MS, Alzheimer's, and COVID-19 datasets. While the MS dataset is used for GA to optimize the CNN hyperparameters and test the optimized CNN, Alzheimer's and COVID-19 are used only for the generalization test of the optimized CNN.

MS is a non-traumatic disease that affects the central nervous system. It has a high potential for disability. The disease diagnosis is based on the analysis of images obtained through magnetic resonance imaging (MRI). The MS dataset consists of four classes: Control-Axial, Control-Sagittal, MS-Axial, and MS-Sagittal, which contain 3427 images. The images in the dataset were preprocessed by resizing them to 150×150 pixels to ensure consistency with the model input. The processed images and their corresponding labels were then prepared for model training. The dataset can be downloaded at: <https://www.kaggle.com/datasets/buraktaci/multiple-sclerosis>.

The Alzheimer dataset consists of MRI images. It includes four classes: Mild Dementia, Moderate Dementia, No Dementia, and Very Mild Dementia. The dataset comprises 8960 images for Mild Dementia, 6464 images for Moderate Dementia, 9600 images for No Dementia, and 8960 images for Very Mild Dementia. In this study, several preprocessing steps were applied to the Alzheimer's image dataset to enhance model performance. The images were resized and normalized to ensure consistency in input format. To address class imbalance and reduce overfitting, data augmentation techniques and the Synthetic Minority Over-sampling Technique algorithm were utilized. These steps resulted in a balanced and diverse dataset suitable for training. The dataset can be downloaded at: <https://www.kaggle.com/datasets/uraninjo/augmented-alzheimer-mri-dataset/data>.

COVID-19 is a virus that has led to severe infections, resulting in significant mortality. Radiological imaging of the chest has been one of the fastest and most accurate methods for COVID-19 diagnosis. The COVID-19 dataset consists of chest X-ray images and includes three classes: COVID-19, Normal, and Pneumonia. It contains 5228

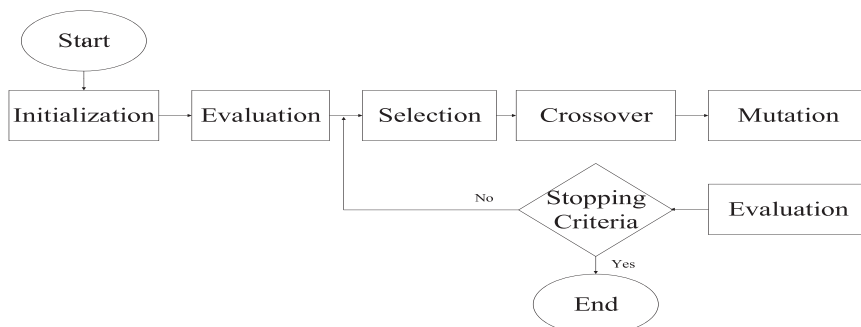


Fig. 2. A Simple GA steps.

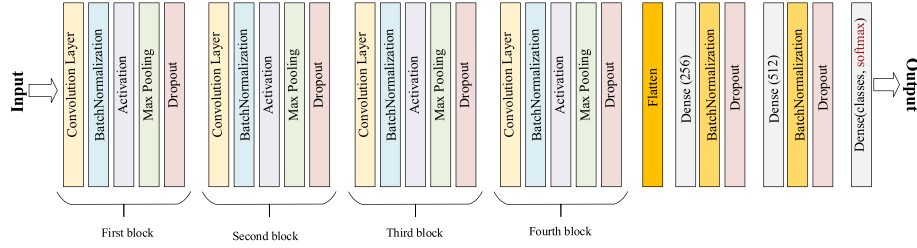


Fig. 3. An example CNN architecture.

images [44]. In this dataset, class labels were encoded as numerical values, and the images were resized to ensure uniform input dimensions. The data was then prepared for model training. The dataset is publicly available at: <https://www.kaggle.com/datasets/sachinkumar413/covid-pneumonia-normal-chest-xray-images/data>.

By default, these datasets are not split into training, testing, and validation sets. 20% of each dataset is split into the test set; 20% of the remaining images are split into the validation set; and the remaining images, 64% of the total images, are used as the training set.

4. Optimization of CNN hyperparameters

4.1. CNN architecture used

Optimizing both the network architecture and hyperparameters simultaneously would require an extremely high computational cost. To make the optimization process feasible, this study focuses on optimizing a set of critical hyperparameters on a fixed CNN architecture with relatively low complexity. The adopted CNN consists of four convolutional blocks followed by a fully connected block, as illustrated in Fig. 3.

Each convolutional block includes a 2D convolution layer followed by batch normalization, an activation function, a max-pooling operation, and a dropout layer. The fully connected block is composed of two hidden dense layers and one output classification layer, where batch normalization and dropout are applied after the hidden layers. This design was chosen because it provides a balance between expressive capacity and computational efficiency, making it suitable for hyperparameter optimization using a metaheuristic algorithm.

Prior to fixing this design, preliminary experiments were conducted with three and five convolutional blocks. Architectures with fewer blocks (e.g., three) were found to underfit, especially on the MS dataset, while deeper structures (five blocks) significantly increased the training time without providing meaningful improvements in performance. Based on this empirical observation, a four-block CNN was selected as the baseline architecture to be optimized.

4.2. Hyperparameter selection and genetic algorithm configuration

Convolutional neural networks involve numerous hyperparameters, but optimizing all of them over wide ranges using a GA would be computationally expensive and impractical. Therefore, to ensure efficiency while still improving learning performance, we focused on a subset of hyperparameters with the most significant impact on model behavior. These included the activation function, batch size, pool size, padding type, number of filters, kernel size, and dropout rate. These parameters directly influence non-linearity, regularization, feature extraction capacity, and computational efficiency. Value ranges for each parameter were defined based on standard practices and preliminary empirical evaluations. The GA then explored combinations within these bounds, using the network's training loss on the MS dataset as the fitness function, which enabled systematic identification of configurations that improved model convergence and generalization.

Other hyperparameters were kept fixed to reduce the search space and avoid excessive computational cost. Specifically, the optimizer (Adam) and weight initialization (Glorot) were fixed due to their

robustness and stability in biomedical imaging tasks. Adam optimizer was selected for its adaptive learning rate and proven effectiveness in CNN training, while Glorot initialization was adopted to stabilize weight distribution and prevent vanishing/exploding gradients. In addition, the number of convolutional blocks and overall architecture depth were kept constant to ensure the model remained computationally feasible.

Regarding GA configuration, population size and number of generations were selected to balance search diversity with computational efficiency. Crossover and mutation rates were fine-tuned to maintain genetic diversity while preventing premature convergence. Before fixing these parameters, several preliminary experiments were conducted with varying population sizes and generation counts. The final configuration was selected as the best trade-off between convergence stability, exploration capability, and computational cost. This configuration allowed the GA to effectively explore the search space without excessive runtime.

4.3. Performance evaluation metrics

For the evaluation of the model, weighted accuracy, weighted precision, weighted recall, and weighted F1-score, which are widely used for performance measurement of classification models, are utilized. Accuracy represents the ratio of correctly classified instances to the total number of instances. Precision indicates how accurate or reliable a model's predictions are by measuring the proportion of true positive predictions among all positive predictions made by the algorithm. Recall, also known as sensitivity, represents the ratio of correctly classified positive instances to the total number of actual positive instances. The F1-score, on the other hand, represents the harmonic mean of precision and recall. The mathematical expressions of these metrics are provided in Eqs. (1)–(4), where TP is the number of data points correctly classified as positive, FP is the number of data points incorrectly classified as positive, TN is the number of data points correctly classified as negative, and FN is the number of data points incorrectly classified as negative. These metrics were selected because they are widely used in medical image classification, particularly in cases involving class imbalance. Unlike plain accuracy, they offer a more comprehensive evaluation, making them especially valuable for evaluating multi-class biomedical datasets.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

5. GA system

Fig. 4 illustrates the general framework of the proposed GP system. In Step 1, the initial population is randomly generated, with each chromosome represented as a bitstring. In Step 2, each chromosome is decoded to extract the encoded hyperparameters, and its fitness is evaluated by

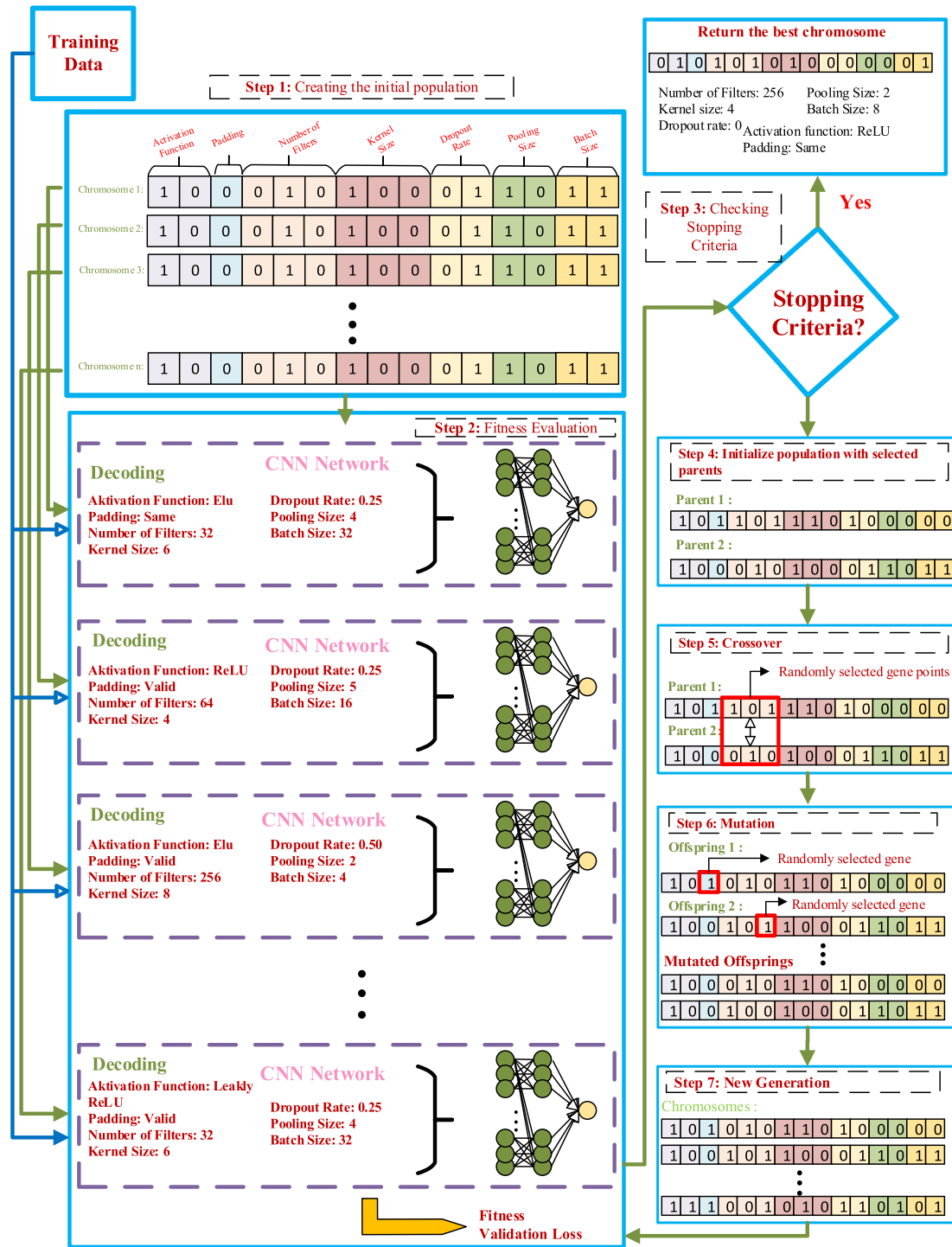


Fig. 4. GA System applied to CNN hyperparameter.

training a CNN using these hyperparameters. Step 3 checks whether the stopping criteria are satisfied; if not, the algorithm iterates through Steps 2–7. In Step 4, parent chromosomes are selected using a selection operator. Step 5 applies a crossover operator to the selected parents to explore promising regions of the search space. In Step 6, mutation is applied to the offspring or parents to maintain genetic diversity. Finally, in Step 7, a new generation is formed and returned to Step 2 for fitness evaluation. This process continues until the stopping criteria are met.

GA is applied to optimize seven crucial CNN parameters: padding, number of filters, kernel size, activation function, dropout rate,

pooling size, and batch size. The optimizer and initializer, which are also important parameters affecting CNN performance, are excluded from the optimization because the search space grows exponentially as more parameters are added to the GA. Instead, these parameters are set to well-known and frequently used ones, namely the Adam optimizer and the Glorot initializer.

GA employs a population of 30 individuals using bit-string encoding, tournament selection with a tournament size of 2, a crossover operator with a probability of 90%, and a mutation operator with a probability of 5%. The system runs for 20 generations.

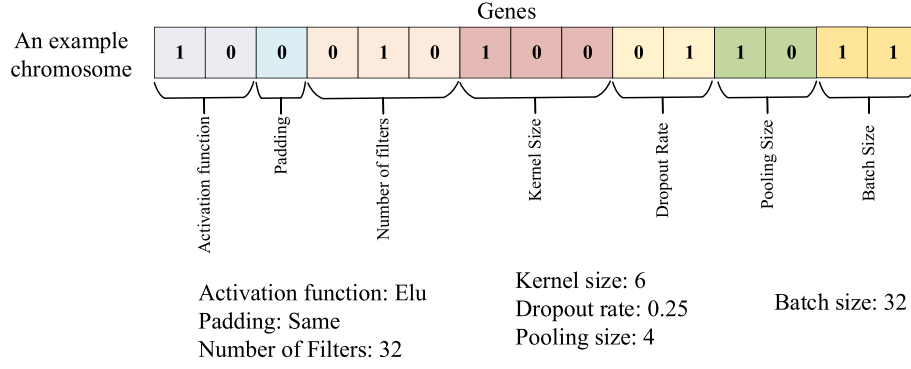


Fig. 5. An example GA chromosome.

The set of possible values that each parameter can take is determined as: activation function = {Tangent hyperbolic, ReLU, Elu, Leaky ReLU}, padding = {Same, Valid}, number of filters = {8, 16, 32, 64, 128, 256, 512, 1024}, kernel size = {2, 3, 4, 5, 6, 7, 8, 9}, dropout rate = {0, 0.25, 0.50, 0.75}, pooling size = {2, 3, 4, 5} and batch size = {4, 8, 16, 32}. The chromosomes are encoded in a binary representation. These parameters are represented by 2, 1, 3, 3, 2, 2, and 2 bits sequentially and respectively; therefore, the total length of the chromosomes is 15. The decimal value of the bit-string corresponding to the parameter indicates the index for the elements of that parameter. Fig. 5 shows an example chromosome with the genes representing the parameters.

Algorithm 1 shows the pseudo-code of our GA system. As mentioned, our system uses a binary encoding representation. It begins by initializing the population randomly. Next, each chromosome in the population is decoded, as shown in Fig. 3, and the corresponding CNN hyperparameters are determined. Then, the chromosome's fitness is measured based on the performance of the CNN using these hyperparameters. After that, the evolution process begins. The population evolves through the generations: it iteratively selects relatively fitter chromosomes, applies crossover and mutation operators to the selected chromosomes,

measures the fitness of the new chromosomes, and replaces the new chromosomes with relatively less fit ones in the population. At the end of the run, the algorithm returns the best chromosome it has ever encountered.

5.1. Fitness evaluation

The fitness of each chromosome is evaluated using the CNN architecture shown in Fig. 3 and the hyperparameters encoded within that chromosome. The resulting CNN is trained for 10 epochs on the MS dataset, with 20% of the training data reserved as a validation set. At the end of training, the validation loss is assigned as the fitness value of the corresponding chromosome.

6. Experimental results

6.1. Results of GA application

We conducted multiple attempts to optimize the CNN hyperparameters using the GA, with the settings described in the previous section, applied to the CNN architecture illustrated in Fig. 3. While the GA consistently identified hyperparameter configurations that yielded high accuracy, some of these configurations failed to generalize to different datasets. Consequently, we selected the optimal configuration by prioritizing its generalization performance over performance on a single dataset. The most effective chromosome obtained from these GA runs, along with the corresponding optimized hyperparameters, is presented in Fig. 6. As illustrated in the figure, our GA system selected the following optimal hyperparameters: ReLU as the activation function, 'same' padding, 256 filters, kernel size 4, dropout rate 0, pooling size 2, and batch size 8. Since the optimal dropout rate is 0, dropout layers are not utilized.

6.2. Performance and generalization evaluation of the optimized hyperparameter

The CNN model, whose hyperparameters are optimized through GA, is evaluated on the MS, Alzheimer's, and COVID-19 datasets. The MS dataset is used both during the hyperparameter evolution phase and to assess the performance of the evolved model. In contrast, the Alzheimer's and COVID-19 datasets are excluded from the GA process to ensure a fair comparison and to evaluate the generalization capability of the evolved model.

Table 2 shows the results of classification performance metrics obtained on the MS, Alzheimer, and COVID-19 datasets using the comparison models employed in the study, including Xception, VGG16, AlexNet, ResNet50, InceptionV3, VGG19, GoogleNet, and the evolved model. In the MS dataset, the proposed model exhibits superior performance across all metrics (96.81% precision, 94.96% recall, 95.65% F1-score, and 96.06% accuracy). InceptionV3 demonstrates the highest success among the comparison models. ResNet50 is the model with

Algorithm 1 Pseudo-code showing the hyperparameter optimization of CNN using GA.

```

1: Begin: Generate a population of chromosomes that consist of bit-string.
2: for each chromosome in population do
3:   Decode the chromosome and set corresponding hyperparameters of CNN to the decoded parameters
4:   Measure the fitness of the chromosome
5: end for
6: for generation = 1 to maximum number of generations do
7:   for iteration = 1 to population size/2 do
8:     Apply the selection operator to the population to select two chromosomes as the parents
9:     if the crossover probability is met then
10:      Apply the crossover operator to the parents and create two offspring
11:     end if
12:     Apply the mutation operator to the new chromosomes (parents or offspring)
13:     Decode the new chromosomes and set corresponding hyperparameters of CNN to the decoded parameters
14:     Measure the fitness of the new chromosomes
15:   end for
16: end for
17: return: The chromosome that has the highest fitness value throughout the generations

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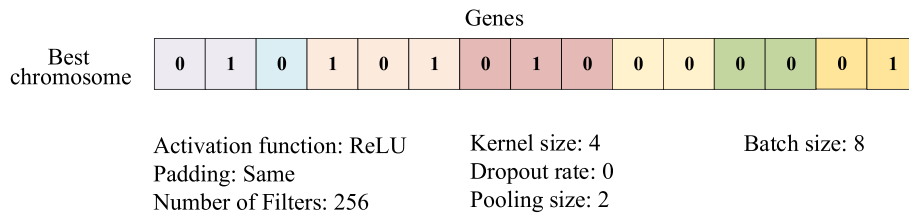


Fig. 6. The chromosome optimized by GA and hyperparameters decoded from the chromosome.

Table 2

Precision, recall, F1-score and accuracy of the various state-of-the-art models and the proposed model on the testing sets of Alzheimer's, COVID-19 and MS datasets.

Dataset	Models	Precision	Recall	F1-Score	Accuracy
MS	Xception	0.8734	0.8601	0.8607	0.8323
	VGG16	0.8356	0.8280	0.8201	0.8279
	Alexnet	0.8168	0.8178	0.8126	0.8177
	Resnet50	0.7163	0.7187	0.6949	0.7186
	InceptionV3	0.8658	0.8659	0.8649	0.8658
	VGG19	0.8534	0.8542	0.8534	0.8542
	GoogleNet	0.8268	0.8032	0.8059	0.8032
	Proposed Model	0.9681	0.9496	0.9565	0.9606
Alzheimer	Xception	0.7941	0.7939	0.7884	0.7939
	VGG16	0.7806	0.7715	0.7712	0.7714
	Alexnet	0.8074	0.8042	0.8038	0.8041
	Resnet50	0.7267	0.7275	0.7264	0.7275
	InceptionV3	0.9147	0.9141	0.9142	0.9140
	VGG19	0.8773	0.8779	0.8774	0.8779
	GoogleNet	0.8854	0.8848	0.8850	0.8847
	Proposed Model	0.9650	0.9644	0.9645	0.9643
COVID-19	Xception	0.9404	0.9388	0.9387	0.9388
	VGG16	0.9379	0.9379	0.9378	0.9378
	Alexnet	0.9303	0.9302	0.9298	0.9302
	Resnet50	0.8602	0.8499	0.8443	0.8450
	InceptionV3	0.9257	0.9235	0.9237	0.9235
	VGG19	0.9175	0.9149	0.9156	0.9149
	GoogleNet	0.9327	0.9207	0.9214	0.9206
	Proposed Model	0.9494	0.9465	0.9463	0.9464

the lowest success on the MS dataset. A similar trend is observed in the Alzheimer's and COVID-19 datasets, where the proposed model outperformed all other models and achieved the highest scores in all metrics. While InceptionV3 follows the proposed model and achieves higher performance than the other comparison models, the ResNet50 demonstrates the lowest performance on the Alzheimer's and COVID-19 datasets.

In summary, the proposed model, evolved by the GA, outperforms other state-of-the-art models not only on the dataset used during GA evolution but also on the datasets not used during the GA evolution phase. InceptionV3 exhibits the most competitive overall performance among the ready-made models. On the other hand, ResNet50 performs worse than other models on the datasets used.

Fig. 7 shows the comparison results of the model generated by the GA with the other well-known and commonly used CNN models based on validation accuracy obtained using three datasets (MS, COVID-19, and Alzheimer's). The evolved model consistently achieves superior validation accuracy compared to other widely used state-of-the-art deep learning architectures under the same training conditions. On the MS dataset, the proposed model again exhibits strong performance with an accuracy of around 96%, while the other networks remain below 87%, and the average gain in this dataset is 17%. Similarly, on the Alzheimer's dataset, the proposed model achieves a peak validation accuracy of approximately 97%, outperforming the compared models, which achieve between 73% and 93%, indicating a 4–34% performance gain. For the COVID-19 dataset, the proposed model achieves the highest accuracy, reaching nearly 95%, followed by Xception, VGG16, AlexNet, InceptionV3, GoogleNet and VGG19, which reach values varying

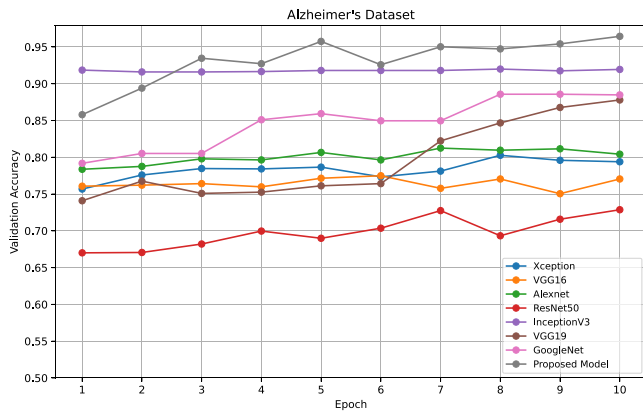
between 92%, and 93%. ResNet50 remains far below the others, reaching 85% accuracy. The validation accuracy curves across all datasets show that the evolved model converges faster and maintains greater stability and generalization capability.

These results, shown in Table 2 and Fig. 7, highlight the optimized CNN's robustness and enhanced learning efficiency, which consistently outperforms state-of-the-art architectures in classification accuracy across diverse medical image classification tasks.

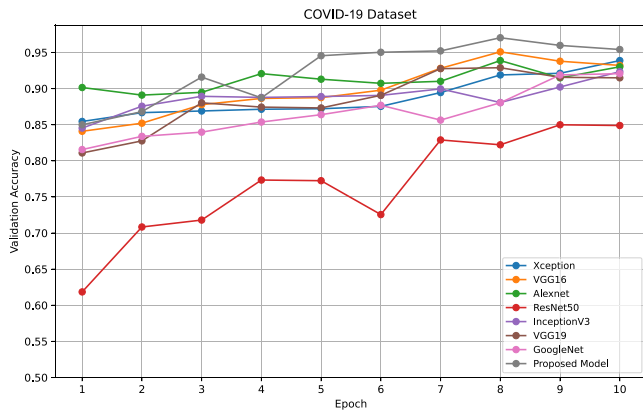
6.3. Sensitivity analysis of the GA-optimized hyperparameters

To assess the robustness of the proposed model and evaluate the influence of each hyperparameter on the overall performance, a sensitivity analysis was conducted on the CNN configuration optimized by the GA. In this analysis, each hyperparameter was individually modified while keeping the remaining parameters fixed at their optimal GA-selected values. The purpose of this test was to observe the model's sensitivity to slight deviations from the optimal configuration and to verify the stability of the optimization results.

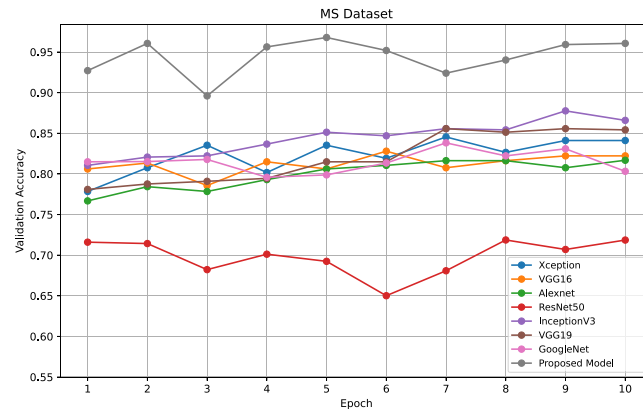
The GA-determined optimal hyperparameter configuration is summarized as follows: activation function = ReLU, padding = Same, number of filters = 256, kernel size = 4, dropout rate = 0, pooling size = 2, and batch size = 8. Each parameter was changed to a nearby alternative value (e.g., increasing the dropout rate to 0.25, using a kernel size of 5 instead of 4, or changing the activation function to ELU). The model was retrained for each modified configuration under the same experimental conditions, and the resulting performance metrics are presented in Table 3.



(a) Alzheimer's dataset



(b) COVID-19 dataset



(c) MS dataset

Fig. 7. Validation accuracy of the various state-of-the-art models and the evolved model on the validation sets of Alzheimer's, COVID-19 and MS datasets.

The results indicate that the CNN performance gradually degrades when any of the optimized hyperparameters is altered. The most notable performance drop occurred when the kernel size or batch size was changed, confirming their strong influence on model generalization and convergence behavior. Minor adjustments in dropout rate or padding type led to smaller degradations, suggesting relatively lower sensitivity to these parameters. Overall, the GA-selected configuration achieved the highest and most balanced performance across all evaluation metrics

Table 3

Sensitivity analysis of the proposed CNN with respect to individual hyperparameters.

Configuration (Modified parameter)	Precision	Recall	F1-Score	Accuracy
Dropout Rate = 0.25	0.9428	0.9409	0.9408	0.9409
Kernel Size = 5	0.9073	0.8652	0.8651	0.8652
Number of Filters = 128	0.9358	0.9351	0.9349	0.9350
Padding = Valid	0.9455	0.9458	0.9456	0.9458
Pooling Size = 4	0.9204	0.9214	0.9203	0.9213
Batch Size = 32	0.8910	0.8330	0.8328	0.8330
Activation Function = ELU	0.9281	0.9272	0.9265	0.9272
GA-Optimized Parameters	0.9650	0.9644	0.9645	0.9643

Table 4

Performance comparison of the proposed GA-optimized model and benchmark CNN architectures on the CIFAR-10 dataset.

Model	Precision	Recall	F1-score	Accuracy
Xception	0.7750	0.7724	0.7717	0.7724
VGG16	0.7226	0.7163	0.7177	0.7163
AlexNet	0.7034	0.6985	0.6944	0.6985
ResNet50	0.8210	0.8185	0.8177	0.8185
InceptionV3	0.6711	0.6688	0.6690	0.6688
VGG19	0.7806	0.7806	0.7796	0.7806
GoogleNet	0.6667	0.6653	0.6646	0.6653
Proposed Model	0.8248	0.8098	0.8092	0.8098

(precision, recall, F1-score, and accuracy), demonstrating the stability and reliability of the genetic optimization process.

6.4. Cross-domain evaluation on CIFAR-10 dataset

To further assess the generalization ability of the GA-optimized CNN architecture beyond the biomedical imaging domain, an additional evaluation was conducted on the CIFAR-10 dataset [45], which contains 60,000 color images (32×32 pixels) distributed equally across 10 object categories. The dataset was divided into 50,000 images for training and 10,000 for testing. Each model was trained for 10 epochs under identical experimental conditions. While the comparative deep networks (Xception, VGG16, AlexNet, ResNet50, InceptionV3, VGG19, and GoogleNet) utilized transfer learning with pretrained weights, the proposed GA-optimized CNN was trained from scratch using the same data splits.

As shown in Table 4, the proposed GA-optimized CNN achieved an accuracy of 80.98%, performing comparably to deeper architectures such as ResNet50 (81.85%) and surpassing several widely used models, including VGG16, AlexNet, and InceptionV3. Although the architecture was primarily optimized for biomedical image classification, it still demonstrated competitive and in several cases superior performance on the general-purpose CIFAR-10 dataset. This result highlights the robustness and adaptability of the GA-based optimization process, indicating that the evolved model parameters capture generalizable visual patterns rather than domain-specific features. Therefore, the proposed model can be considered a compact yet effective alternative to more complex CNN architectures when cross-domain generalization is desired.

6.5. Discussion

The findings of this study demonstrate that the CNN model optimized through GA consistently outperformed state-of-the-art deep learning architectures across three distinct medical imaging datasets. On the MS dataset, which was used during GA evolution, the proposed model achieved the highest precision, recall, F1-score, and accuracy. More importantly, the optimized model also generalized effectively to independent datasets (Alzheimer's and COVID-19), confirming that the GA did not merely overfit to the optimization dataset but instead identified robust and transferable hyperparameter configurations.

Furthermore, when evaluated on a non-biomedical dataset (CIFAR-10), the model maintained competitive performance, further validating its cross-domain generalization capability.

When compared with widely used models such as VGG16, VGG19, AlexNet, GoogleNet, ResNet50, InceptionV3, and Xception, the GA-evolved CNN achieved superior performance across all evaluation metrics on biomedical datasets. Among the baselines, InceptionV3 yielded the most competitive results, while ResNet50 consistently showed the lowest scores in biomedical experiments. However, in the cross-domain evaluation on the CIFAR-10 dataset, ResNet50 achieved the best overall accuracy among the baseline models, whereas the proposed GA-optimized CNN produced comparable results, demonstrating its strong generalization capability beyond the biomedical domain. These outcomes highlight the advantage of GA-based optimization over conventional architectures and transfer learning strategies, further reinforcing its potential in medical image classification tasks.

The improved generalization ability of the evolved CNN suggests that systematic hyperparameter optimization contributes not only to dataset-specific performance but also to cross-domain adaptability. The observed faster convergence and stable training dynamics (Tables 2 and 4) support this explanation, indicating that GA-based optimization enhances representational efficiency and reduces the risk of overfitting. This is particularly significant for medical imaging, where robustness across disease types and modalities is essential.

A notable strength of the proposed method is its ability to reduce reliance on manual trial-and-error hyperparameter tuning by providing an automated and systematic optimization framework. At the same time, its consistent success on multiple datasets underscores the clinical relevance of the approach. Nevertheless, the method is computationally demanding, as GA requires multiple training runs to evaluate candidate solutions. While manageable within the scope of this study, scaling the approach to more complex architectures or larger datasets may require additional computational resources.

In conclusion, GA-driven hyperparameter optimization enhances CNN performance and generalization in biomedical imaging, consistently outperforming well-established deep learning models. Future research could expand on this work by incorporating a broader set of hyperparameters, exploring alternative evolutionary strategies, or extending the approach to non-biomedical applications such as video analysis or segmentation. Such directions would further validate the versatility and applicability of evolutionary optimization in deep learning.

7. Conclusion

In this study, a CNN optimized using GA achieves superior performance in biomedical image classification problems. Experimental findings show that the GA-optimized CNN model classifies images with higher performance compared to well-known deep networks such as Xception, VGG16, AlexNet, ResNet50, InceptionV3, VGG19, and GoogleNet across different datasets used in MS, Alzheimer's, and COVID-19 classification. The evolved CNN model not only achieves the best results on MS brain MRI images but also demonstrates superior consistent performance on the Alzheimer's MRI and COVID-19 X-ray datasets, despite not being trained on these datasets. This outcome indicates that the optimized hyperparameter set remains effective across different biomedical imaging conditions, highlighting the model's strong generalization ability. Notably, the fact that the optimal hyperparameters learned on a single dataset (MS) also perform successfully on two other independent datasets without the need for transfer learning further evidences the generalization capability of the proposed approach. This finding suggests that GA-based optimization helps mitigate hyperparameter overfitting, a problem that frequently arises in deep learning models.

As a result, the model achieves superior performance compared to standard architectures on both the MS dataset and the other biomedical test datasets. However, the method also exhibits certain limitations. In the current study, training was limited to 10 epochs to maintain

reasonable computational time; nonetheless, the duration of GA optimization may increase for more complex datasets or broader hyperparameter spaces. To further assess the generalization capability of the optimized CNN, the model was additionally tested on the CIFAR-10 natural image dataset. The results showed that, despite being optimized for biomedical data, the model achieved a competitive accuracy close to ResNet50 and outperformed several other benchmark architectures. This finding indicates that the GA-optimized hyperparameter configuration is not domain-specific and can generalize well beyond medical imaging. Nevertheless, performance consistency across other modalities or tasks (e.g., video analysis or segmentation) still requires further investigation in future studies.

In future work, the proposed GA-based optimization approach can be extended in several directions. First, by adopting a multi-objective optimization framework, the hyperparameter search can consider not only accuracy but also other criteria such as computational complexity or memory efficiency. In this way, both high-performing and resource-efficient CNN models can be designed. Secondly, the proposed method can be expanded to a full neural network architecture search, evolving not only the hyperparameters but also the types and numbers of layers. Such an automatic architecture discovery framework may significantly reduce the difficulty of manual architecture design.

CRediT authorship contribution statement

Ahmet Yilmaz: Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Methodology, Conceptualization. **İlya Kuş:** Writing – original draft, Visualization, Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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